Kinds of municipal zone in Colombia: drinking water, health and education

A descriptive analysis, 2016–2023

2026-11-01

Summary

k-means on nine water, service and context features finds 4 kinds of municipal zone among 11,235 zone-years from 1,090 municipalities, 2016–2023. The lowest mean infant mortality is in "Cities and large towns, no-risk water" and the highest in "Highland towns, no-risk water". Within each kind, the rank correlation between utility-supplied IRCA and infant mortality lies between −0.04 and +0.10. All comparisons are descriptive and do not establish causal effects of water quality.

Question

Which kinds of municipal zone exist in Colombia, in terms of drinking-water quality, services and context, and how do infant mortality, Saber 11 scores and school dropout differ between them?

Data

One row is a municipality's urban or rural zone in one year (`data/water_municipal.csv`, real public data joined on DANE DIVIPOLA codes; see `data/README.md`). **IRCA is the utility-supplied ("prestadores") IRCA** from INS open data (datos.gov.co `nxt2-39c3`), not INS's national total, which also covers self-supplied systems. Coverage comes from REC self-reports, poverty (IPM) from the 2018 census, altitude is a derived SRTM point and SGP transfers are nominal pesos.

Method

We keep municipality zone-years 2016–2023 with an IRCA value (11,235 rows; 6,411 without IRCA dropped). Capitals' urban rows use the municipal coverage total (214 rows, flagged); remaining gaps take the year × zone median (653 rows, flagged). Each feature is z-scored within its year. k-means (20 starts, seed 42) was fitted for k = 2–8; k = 4: the next cluster cuts inertia by only 7% (< 10%), and its silhouette 0.195 is within 0.01 of the best k >= 3 (0.196). Clusters are numbered by centroid IRCA and named by a fixed rule. Outcomes were never used to form the clusters.

What sets each kind apart

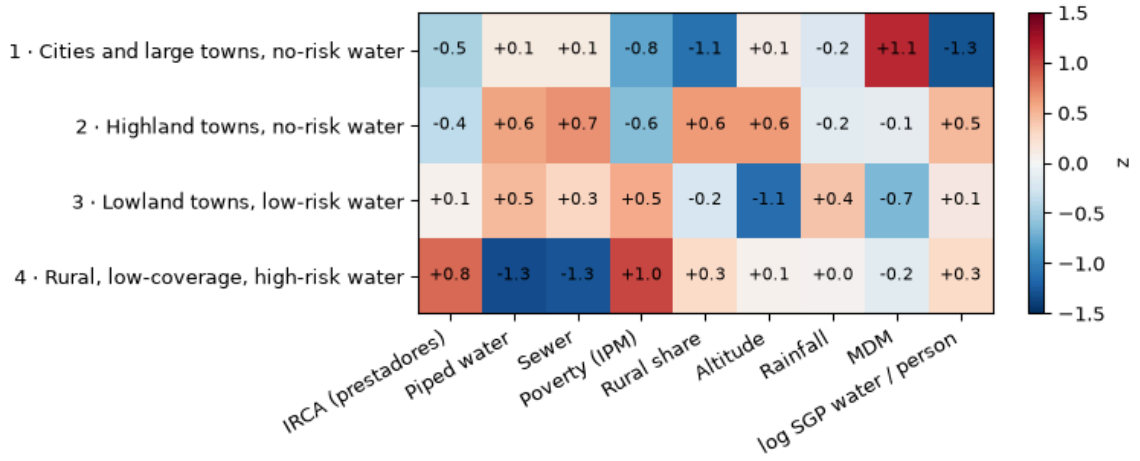


Figure 1: Cluster mean z-scores: standard deviations above (red) or below (blue) the year's average. IRCA is utility-supplied (prestadores).

Table 1: The kinds: zone-years, municipalities and median utility-supplied (prestadores) IRCA.

cluster	kind	zone-years	municipalities	median IRCA
1	Cities and large towns, no-risk water	2271	284	1.6
2	Highland towns, no-risk water	3750	558	4
3	Lowland towns, low-risk water	2451	479	9.2
4	Rural, low-coverage, high-risk water	2763	729	35.2

Outcomes by kind

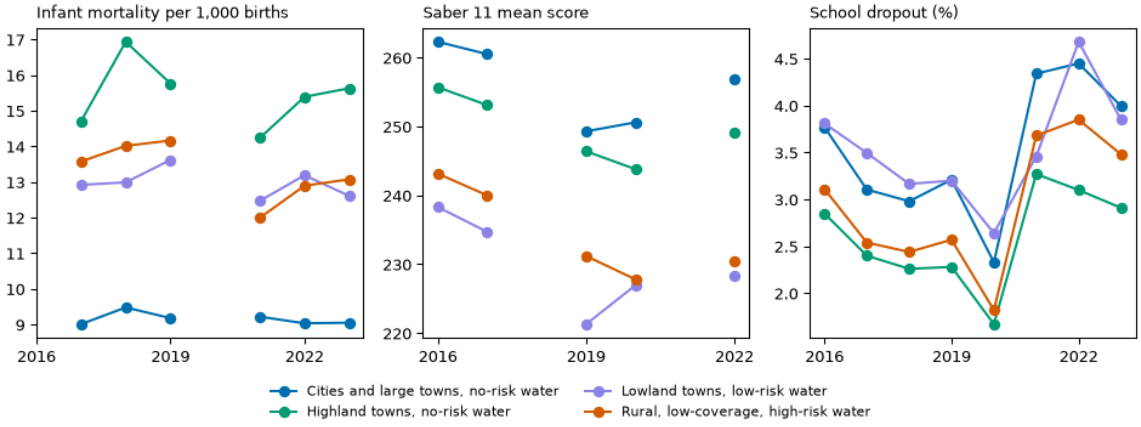


Figure 2: Median outcome by cluster and year. Infant mortality: 2017–2019 and 2021–2023 only (method change in 2017; 2020 looks like another source). Saber 11: published years. Gaps are years without data.

Correlation ≠ causation

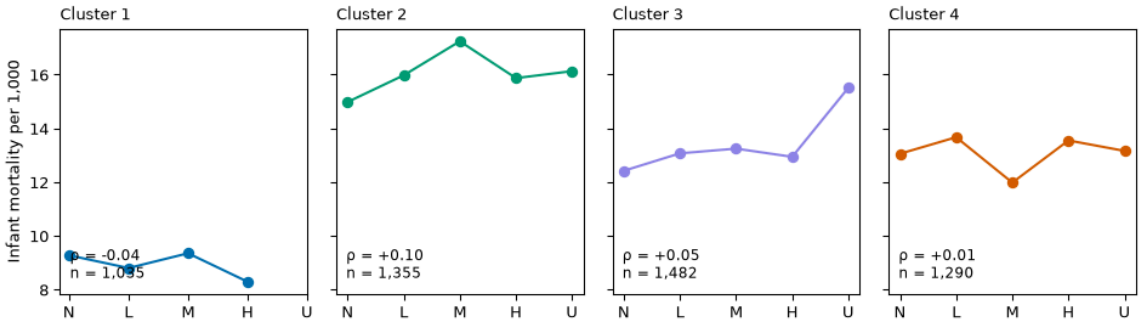


Figure 3: Median infant mortality by utility-supplied (prestadores) IRCA band within each cluster, 2017–2019 and 2021–2023 pooled; ρ is Spearman's rank correlation, n municipality-years.

Within each kind, infant mortality hardly changes with the IRCA band (ρ between −0.04 and +0.10). That does not show water is harmless, nor that it is harmful: poverty, rurality, access to health services and reporting quality affect both water and recorded deaths, and infant mortality is municipal while IRCA is zonal (an ecological comparison). These differences are descriptive and do not establish causal effects of water quality.

Limits

- IRCA is utility-supplied only; self-supplied rural systems, with higher risk, are not in it, and rural IRCA is missing in about half of rural rows.
- Infant mortality changes method in 2017; small municipalities have noisy rates, and a blank may mean no recorded deaths.
- Saber 11 covers students who sat the exam, by school location; 2018 and 2021 are missing.
- Silhouette 0.19: the kinds overlap and summarise a continuum.

Sources

- INS, *Calidad del Agua para Consumo Humano en Colombia*, datos.gov.co `nxt2-39c3` (CC BY-SA 4.0), utility-supplied (prestadores) IRCA.
- Resolución 2115 de 2007, art. 15 (IRCA risk bands).
- DNP TerriData (coverage, IPM, MDM, SGP, infant mortality; Superservicios, DANE, DNP-SICODIS, MinSalud).
- DANE DIVIPOLA `gdxc-w37w`, population projections and MGN 2023.
- ICFES Saber 11 (`kgxf-xxbe`, `rnvb-vnyh`); MEN dropout (`nudc-7mev`).
- CHIRPS v3.0 rainfall (Funk et al. 2026, doi:10.1038/s41597-026-07096-4); NASA SRTM GL3 via OpenTopoData (doi:10.5067/MEaSUREs/SRTM/SRTMGL3.003).

Full details, licences and retrieval dates: `data/README.md`.